MODULE 2

- **Depth Estimation And Multi-Camera Views:**

  Depth Estimation and Multi-Camera Views: Perspective, Binocular Stereopsis: Camera and Epipolar Geometry; Homography, Rectification, DLT, RANSAC, 3-D reconstruction framework; Auto-calibration.

https://www.slideshare.net/slideshow/module-2-computer-vision-part-2-depth-estimation/265230286#16

Depth Estimation

Depth Estimation is the task of measuring the distance of each pixel relativ the camera. Depth is extracted from either monocular (single) or stereo (multi views of a scene) images. Traditional methods use multi-view geometry to the relationship between the images. Newer methods can directly estimate depth by minimizing the regression loss, or by learning to generate a novel view from a sequence.

It is an important task in computer vision and has various applications such as 3D reconstruction, augmented reality, autonomous navigation, and more.

There are several techniques for depth estimation, and one commonly used approach is stereo vision. Stereo vision involves using a pair of cameras, known as a stereo camera setup, to capture images of a scene from slightly different viewpoints. The disparity between corresponding pixels in the left and right images can be used to calculate the depth information.

Depth Estimation

- To estimate depth using stereo vision, the following steps are typically involv

Camera calibration: Accurate calibration of the stereo camera setup is necessar determine the intrinsic and extrinsic parameters of each camera. This calibrati process establishes the relationship between the 3D world coordinates and the corresponding 2D image points.

Image rectification: Rectification is performed to transform the stereo image pair so that corresponding epipolar lines become scanlines. This simplifies the matching process by reducing it to a 1D search problem.

Disparity calculation: Matching algorithms are used to find correspondences between the left and right images. These algorithms aim to identify the pixel disparities, i.e., the horizontal shift of a point between the two images. Common techniques include block matching, semi-global matching, and graph cuts.

Depth computation: Once the disparity map is obtained, the depth can be calcul using triangulation. By knowing the baseline distance (distance between the two camera centers) and the focal length of the cameras, the depth at each pixel can computed using simple geometry.

Apart from stereo vision, there are other methods for depth estimation, including structured light, time-of-flight, and monocular depth estimation using a single camera. Monocular depth estimation relies on various cues, such as texture, motion, perspective, and object size, to infer depth information. Deep learning-based approaches, especially convolutional neural networks (CNNs), have shown promising results in monocular depth estimation by learning from large-scale datasets.

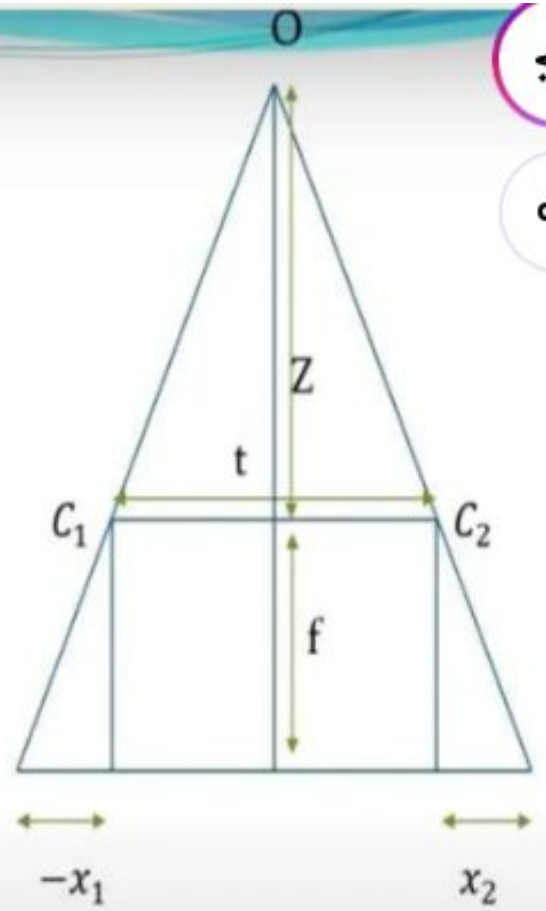
- By similarity of triangles

$$\frac{Z+f}{Z} = \frac{x_2-x_1+t}{t}$$

$$1+\frac{f}{Z} = \frac{x_2-x_1}{t}+1$$

$$Z = \frac{ft}{(x_2-x_1)}$$

$$Z \propto \frac{1}{(x_2-x_1)}$$



Multi-camera views refer to the use of multiple cameras positioned at diffe locations or angles to capture a scene simultaneously. By combining the views f multiple cameras, it becomes possible to obtain a more comprehensive understanding of the scene, including depth information and different perspective Here are some key points about multi-camera views:

Enhanced Coverage: With multiple cameras, it is possible to cover a larger area of the scene compared to a single camera. Each camera can capture a different portion or angle of the scene, providing a wider field of view.

Improved Depth Perception: By utilizing multiple cameras, depth information can be extracted using techniques like stereo vision or structure from motion. By comparing the views from different cameras, it becomes possible to estimate the depth of objects in the scene, enabling 3D reconstruction and depth-based applications.

Redundancy and Robustness: Having multiple camera views provides redundancy in capturing the scene. If one camera fails or its view is obstructed, other cameras can still provide information about the scene. This redundancy enhances the robustness and reliability of the system.

Viewpoint Diversity: Each camera in a multi-camera setup can have a diffe perspective or viewpoint of the scene. This diversity of viewpoints can be beneficial for various applications, such as object tracking, activity recognition, or sc understanding. By combining different perspectives, a more comprehensive representation of the scene can be obtained.

Multi-Modal Information: Multi-camera views can also capture different modalities of the scene, such as visible light, infrared, depth sensors, or thermal imaging. By combining these different modalities, richer and more detailed information about the scene can be obtained, leading to improved understanding and analysis.

Applications of multi-camera views include surveillance systems, autonomous vehicles, virtual reality, augmented reality, robotics, sports analysis, and many more. The synchronized and coordinated use of multiple cameras enables a deeper understanding of the scene, enhances accuracy and robustness, and opens up new possibilities in computer vision and imaging applications.

Perspective

Multi-camera views refer to the use of multiple cameras to capture diffe perspectives simultaneously. These multiple camera angles are then often edited together to create a dynamic and engaging visual experience for the audience. E camera provides a unique perspective, allowing viewers to see different angles, details, and reactions.

Multi-camera setups are commonly used in various media productions, including television shows, live events, sports broadcasts, and films. Here are some key perspectives achieved through multi-camera views:

Wide Shots: A wide shot provides an overall view of the scene, capturing the entire set or location. It establishes the context, shows the spatial relationships between characters or objects, and sets the stage for more detailed shots.

Medium Shots: Medium shots focus on characters or objects from a medium distance. They offer a balanced view, showing the subject from the waist up or from the knees up. Medium shots are often used for dialogue scenes and allow viewers to see facial expressions and body language.

Perspective

Close-ups: Close-up shots zoom in on a specific subject, such as a person's face or an object. They highlight details and emotions, creating an intimate connection between the viewer and the subject. Close-ups are particularly effective in conveying emotions or emphasizing important story elements.

Over-the-Shoulder Shots: Over-the-shoulder shots are commonly used in dialogue scenes. They capture the back of one person's shoulder and part of their head, with the main focus on the person they are facing. This perspective provides a sense of depth and helps viewers feel like they are part of the conversation.

Reaction Shots: Reaction shots capture the emotional responses or reactions of characters to a particular event or dialogue. They are usually close-ups of a character's face, emphasizing their expressions and adding depth to the scene.

Point-of-View Shots: Point-of-view shots provide the audience with the perspective of a particular character. The camera becomes the character's eyes, showing what they see and their subjective experience of the situation. These shots can create a sense of immersion and empathy.

Perspective

By combining and switching between these different camera perspectives, direc and editors can create engaging visual narratives that enhance the storytelling experience. Multi-camera views provide flexibility in post-production, allowing the selection of the best shots and angles to convey the intended message and evoke the desired emotions from the audience.

Binocular Stereopsis

Binocular stereopsis is the ability of humans (and some animals) to perceive d and three-dimensional information by utilizing the binocular disparity resulting fro having two eyes placed horizontally on the face. Each eye captures a slig different view of the world, and the brain combines these two images to create a single perception with depth perception.

The process of binocular stereopsis involves several steps:

Binocular Disparity: Binocular disparity refers to the differences in the retinal images between the two eyes. Because the eyes are horizontally separated, they receive slightly different perspectives of the same scene. These disparities are due to the **parallax effect** and provide important depth cues.

- ❑ The parallax effect is a phenomenon that occurs due to the displacement or difference in the apparent position of an object when viewed from different angles. It is a visual cue that helps perceive depth and distance in a scene.

Binocular Stereopsis

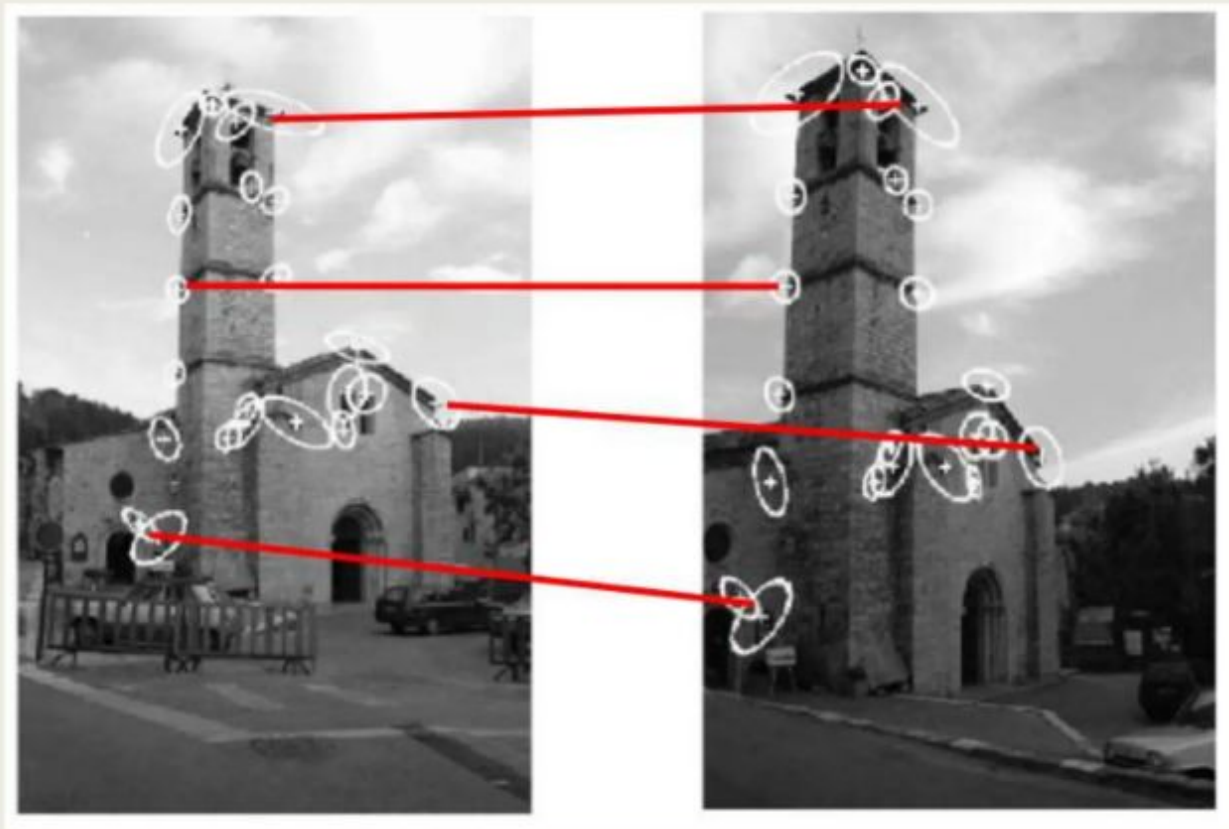
- The parallax effect is closely related to binocular disparity, which is the prim mechanism behind binocular stereopsis (the ability to perceive depth using two eyes). When we view objects with binocular vision, each eye has a slig different perspective, resulting in a disparity between the images captured by each eye. The brain processes these disparities to compute depth information and create a perception of three-dimensional space.

Here's an example to illustrate the parallax effect:

Hold your finger in front of your face and look at it first with your left eye and then with your right eye, alternating between the two. You will notice that the finger appears to shift its position relative to the background. This apparent shift is the parallax effect in action. The amount of shift or displacement is greater when the object is closer to you and smaller when it is farther away.

Binocular Stereopsis

Correspondence Matching: The brain's visual processing system compares images from each eye and matches corresponding points or features between them. It searches for similar patterns, textures, or edges in both images to establ correspondences.



Binocular Stereopsis

Disparity Calculation: Once the corresponding points are identified, the b measures the horizontal displacement or disparity between them. The magnitude of the disparity is proportional to the depth difference between the object and observer.

Depth Perception: By analyzing the magnitude of the disparity, the brain estimates the relative depth of objects in the visual scene. Objects that appear closer will have a larger disparity, while objects farther away will have a smaller disparity.



Binocular Stereopsis

Fusion and 3D Perception: The brain combines the information from both e integrating the two slightly different perspectives into a single perception. This fusion of the images creates the perception of depth, allowing us to see the worl three dimensions.

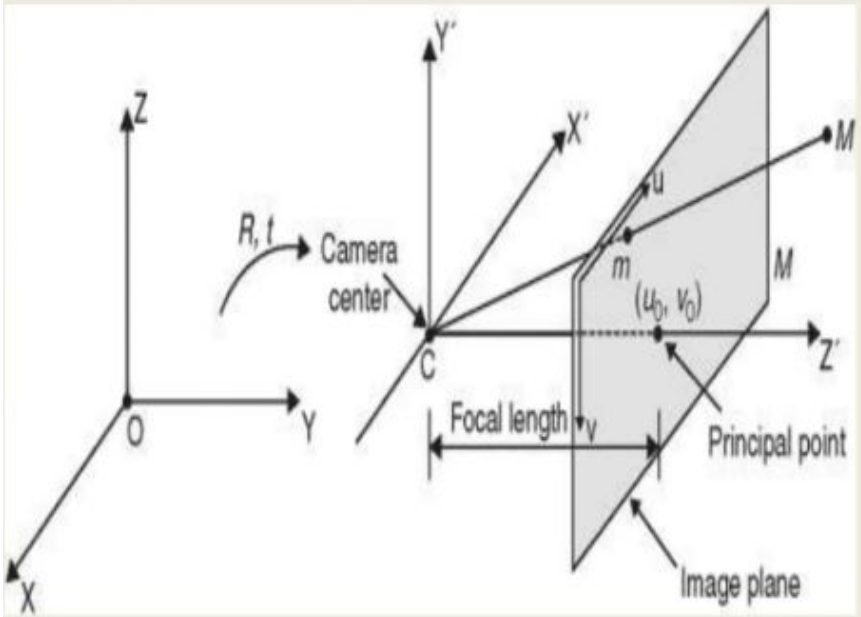
Binocular stereopsis is an important component of human vision and provides us with valuable depth cues, allowing us to navigate and interact with the environment effectively. It enables us to judge distances, perceive the relative positions of objects, and experience a sense of depth and solidity in our visual perception.

In addition to human vision, binocular stereopsis has applications in fields such as computer vision and robotics. By using stereo cameras or other depth-sensing techniques, machines can replicate the principles of binocular stereopsis to perceive depth and reconstruct three-dimensional representations of the world around them.

Camera Geometry

Camera geometry refers to the mathematical and physical properties that describe behavior and characteristics of a camera. It encompasses both intrinsic and extrinsic parameters that define how the camera captures and projects the 3D world onto a image.

Intrinsic Parameters: Intrinsic parameters are internal to the camera and define its internal optical characteristics. These parameters include:

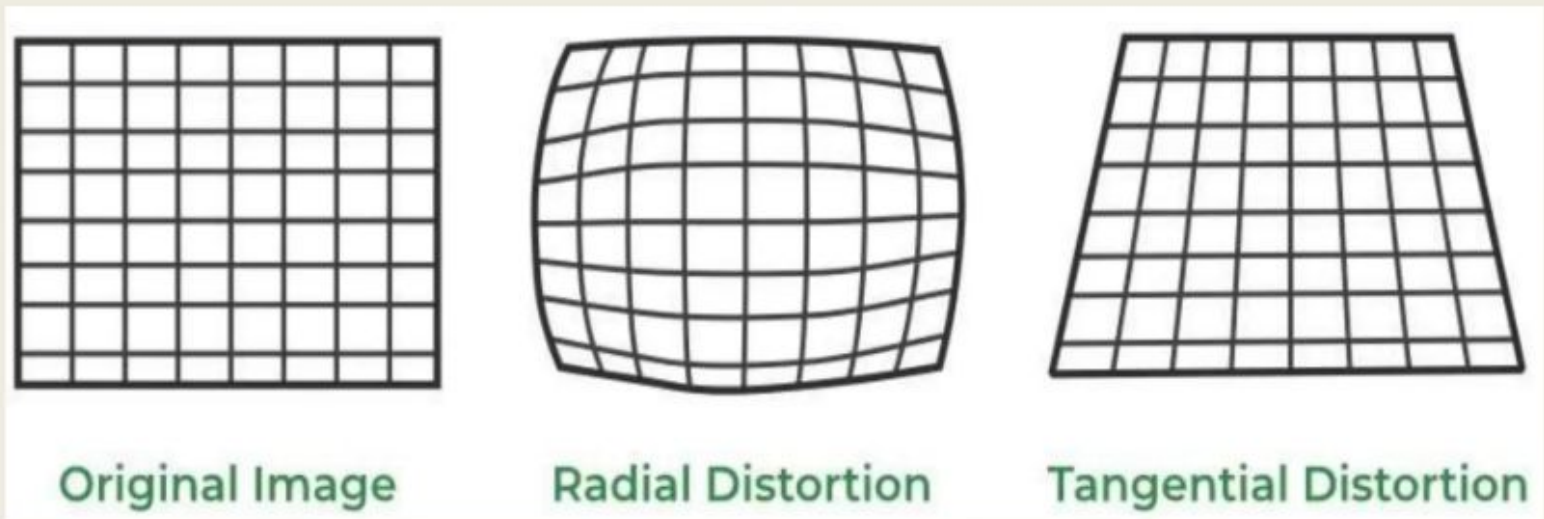


- **Focal Length:** The focal length determines the camera's field of view and the degree of magnification. It represents the distance between the camera's lens and the image sensor when the subject is in focus.
- **Principal Point:** The principal point represents the optical center of the camera. It is the point where the optical axis intersects the image plane.

Camera Geometry

- Lens Distortion: Lens distortion refers to the imperfections in the camera lens that can cause image distortions. Common types of distortion include radial distortion (barrel or pincushion distortion) and tangential distortion.

Tangential Distortion: Tangential distortion is a different type of distortion that occurs due to misalignments or irregularities in the lens elements. It causes the image to appear skewed or stretched asymmetrically, typically in a non-linear manner. *Tangential distortion can result from factors such as slight tilting or displacement of the lens elements or inconsistencies in lens manufacturing.*



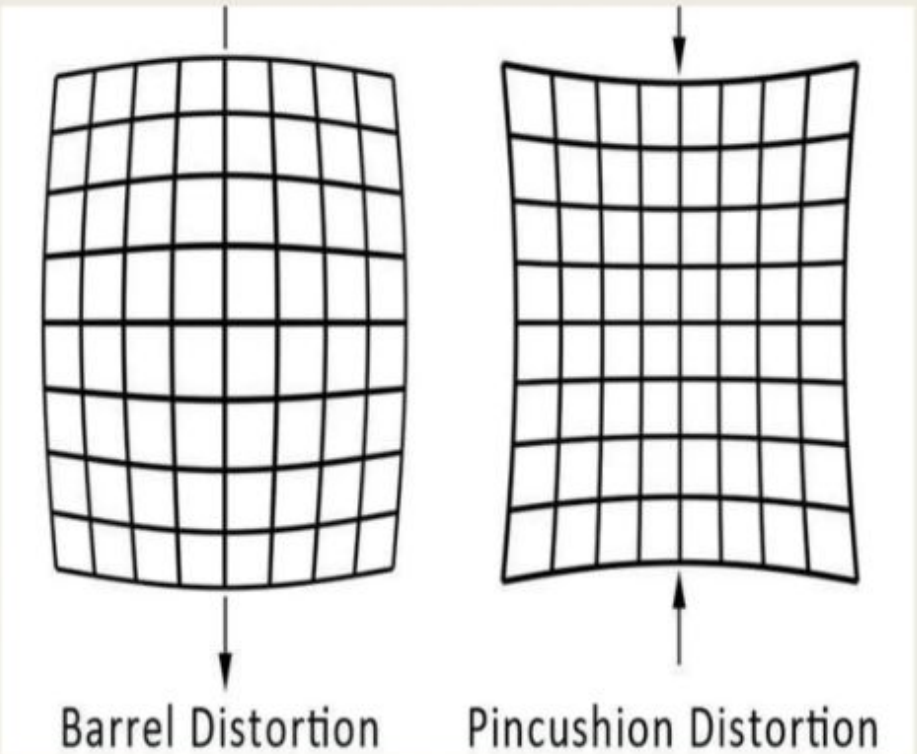
Camera Geometry

Radial Distortion: Radial distortion refers to the distortion that occurs when straight l
near the edges of an image appear curved or bent. *It is caused by imperfections in the lens
cause light rays to refract differently depending on their distance from the center of the lens*

Radial distortion is typically classified into two subtypes:

Barrel Distortion: Barrel distortion causes straight lines to curve outward, resembling the shape of a barrel. It occurs when the outer portions of the image are magnified more than the center. This distortion is commonly observed in wide-angle lenses.

Pincushion Distortion: Pincushion distortion causes straight lines to curve inward, resembling the shape of a pincushion. It occurs when the center of the image is magnified more than the outer portions. Pincushion distortion is often observed in telephoto lenses.



Camera Geometry

Extrinsic Parameters: Extrinsic parameters describe the position and orientation of the camera in the 3D world. These parameters include:

Camera Center: The camera center, also known as the optical center or camera position, represents the location of the camera's optical axis in the 3D world.

Camera Pose: The camera pose describes the position (translation) and orientation (rotation) of the camera relative to a reference coordinate system.

Projection Model: The projection model defines how the 3D world is projected onto the 2D image plane. The most common projection model used is the pinhole camera model, which assumes a perspective projection. It assumes that light rays pass through a single point (pinhole) in the camera and project onto the image plane.

Camera Calibration: Camera calibration is the process of determining the intrinsic and extrinsic parameters of a camera. It involves capturing calibration images with known calibration patterns, such as a chessboard, and using mathematical algorithms to estimate the camera parameters.

Camera Geometry

Understanding camera geometry and its parameters is crucial for var applications, including computer vision, 3D reconstruction, camera calibration, augmented reality, and robotics. By accurately modeling the camera's behavic becomes possible to interpret and manipulate images and accurately estimate the position and geometry of objects in the 3D world.

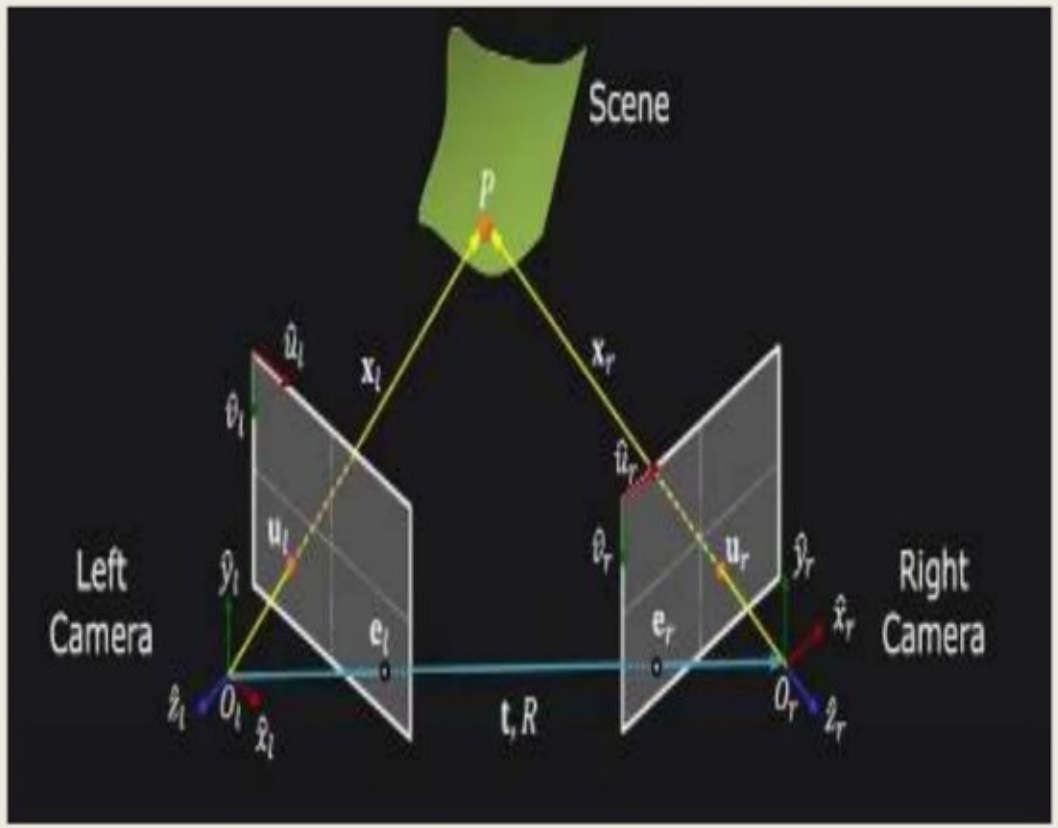
Epipolar Geometry

Epipolar geometry is a fundamental concept in computer vision and stereo ima that describes the geometric relationship between two camera views observing the same scene. It provides constraints on the possible locations of corresponding pc in the two images, enabling depth estimation and 3D reconstruction.

Epipolar geometry include

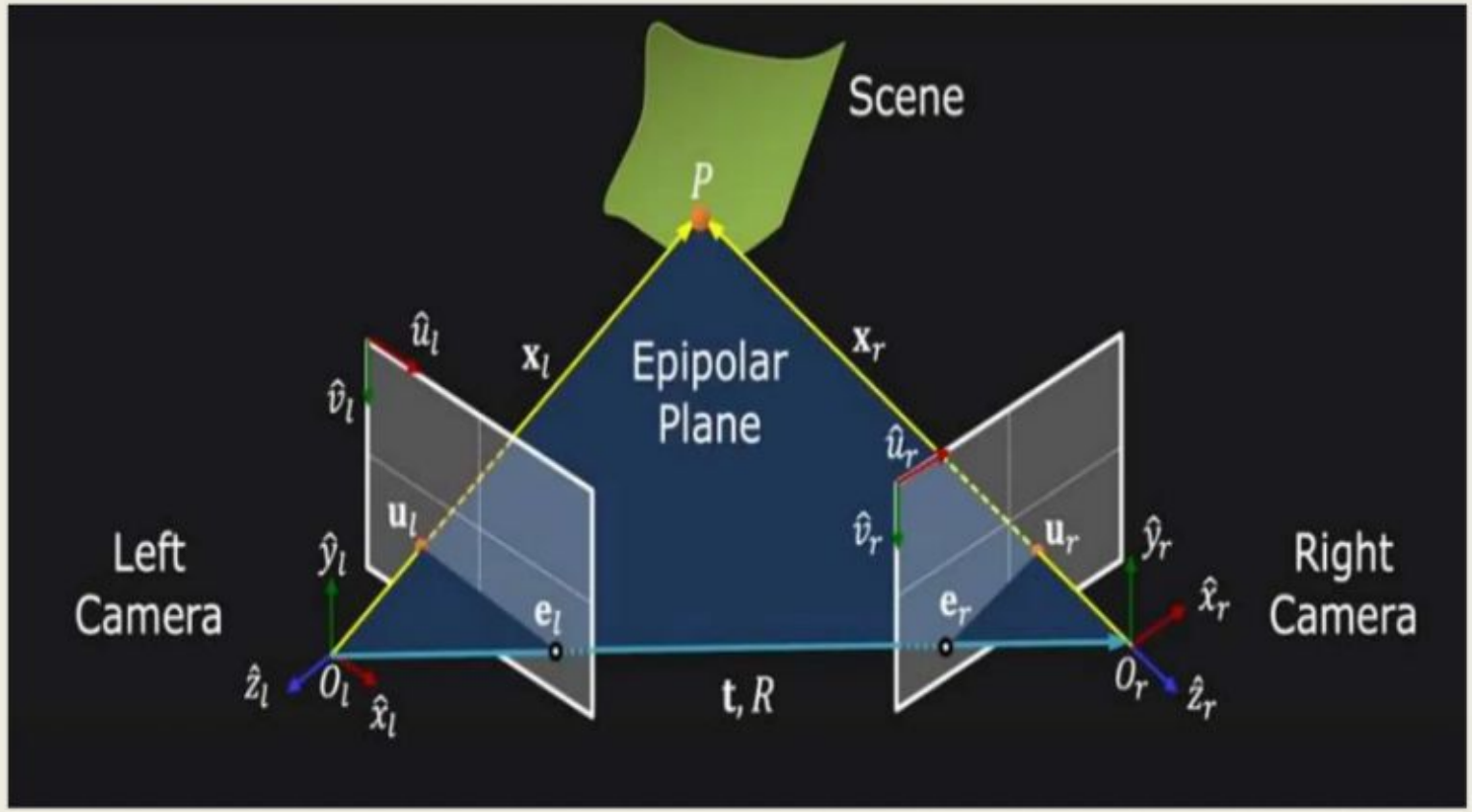
Epipole: The epipole is a point that represents the projection of one camera center onto the image plane of the other camera. It is the point of intersection between the line connecting the camera centers (baseline) and the image plane. Each camera has its own epipole in the other camera's image.e_r

Here e_l and e_r is the epipole



Epipolar Geometry

Epipolar Plane: The epipolar plane is a 3D plane that contains the baseline (the connecting the camera centers) and any point in the 3D scene. It represents the possible locations of corresponding points in the two camera views.



Epipolar Geometry

Epipolar line: The epipolar line is the straight line of intersection of the epip plane with the image plane. It is the image in one camera of a ray through the optical center and image point in the other camera. All epipolar lines intersect at the epip

